AI TRADING AGENTS HACKATHON — LABLAB.AI

CROSSMIND

An auditable capital-protection system
for autonomous trading

Agent #3429 ERC-8004 Identity Registry Built on Kraken CLI

// PROBLEM

THE BLIND SPOT

Every trading bot optimizes for profit.
Nobody asks when the bot should NOT trade.

Most trading systems fail not because they cannot enter trades,
but because they cannot manage loss.

No audit trail. No refusal logic. No survival constraint.

// SOLUTION

SURVIVAL FIRST

> Survival first, profit second — capital preservation is the prime directive
> Risk-based refusals are auditable, not silent failures
> Every decision recorded in a SHA-256 hash-chained Trust Ledger
> AI-powered adversary stress-tests every signal before execution

> *"CrossMind does not just execute trades. It proves why it refused."*

// ARCHITECTURE

4-STAGE PIPELINE

WAR ROOM
Historical stress test

→

GATEKEEPER
Survival scoring

→

LIVE EXEC
Paper trading

→

TRUST LEDGER
Hash-chained audit

4-Agent Architecture

ANALYST
DETERMINISTIC
RSI(14) signals on 4H candles from Kraken OHLC

STRATEGIST
DETERMINISTIC
Position sizing, stop-loss 3%, take-profit 5%

RISK MANAGER
AI-POWERED
Survival scoring, drawdown limits, kill switches

ADVERSARY
AI-POWERED
Red-team attack on every signal before execution

// WAR ROOM

STRESS TEST RESULTS

6 historical crash scenarios replayed from Kraken data

COVID CRASH SURVIVED -24% market, 1.9% DD	LUNA COLLAPSE SURVIVED Signal blocked	FTX CONTAGION SURVIVED Signal blocked
2022 BEAR SURVIVED No entry triggered	FLASH CRASH SURVIVED Signal blocked	RATE SHOCK SURVIVED Signal blocked

6/6
SCENARIOS SURVIVED

1.9%
MAX DRAWDOWN

4
SIGNALS BLOCKED

// TRUST LEDGER

TAMPER-EVIDENT AUDIT TRAIL

SHA-256 hash chain — 5 fields per entry

INTENT	BUY BTC/USD @ $41,230 | RSI=27.4 | Size: 10%
RISK CHECK	Volatility spike detected | Correlation > 0.85
DECISION	BLOCKED — Adversary veto: contagion risk
RESULT	No position opened | Capital preserved
REFUSAL PROOF	Adversary confidence: 0.91 | Kill switch: OFF

prev_hash: a3f8c1...e7d2 → sha256(entry) → next_hash: 7b2e09...f4a1

"CrossMind does not just log trades. It logs *judgment*."

// ON-CHAIN IDENTITY

ERC-8004 INTEGRATION

AGENT ID

#3429

Base Sepolia

IDENTITY NFT

- > On-chain identity for autonomous agent
- > Validation artifacts map to Trust Ledger
- > Immutable record of agent behavior
- > Verifiable by any third party

Every Trust Ledger entry can be cross-referenced with the agent's on-chain identity.

Autonomous trading requires accountable agents.

// TECH STACK

BUILT WITH

Kraken CLI

Paper trading execution
Dead Man's Switch
OHLC market data
MCP integration

ERC-8004 Registry

Agent identity NFT
Base Sepolia deployment
On-chain validation artifacts

Python + Streamlit

Orchestrator pipeline
4-agent architecture
Live monitoring dashboard

SHA-256 Hash Chain

Tamper-evident audit trail
5-field ledger entries
Refusal proof logging

// ACCESS

LINKS

DASHBOARD	huggingface.co/spaces/barbarawu/crossmind-dashboard
GITHUB	github.com/yaowubarbara/crossmind
ERC-8004 TX	sepolia.basescan.org/tx/0x5ab87e...074d7e6

CROSSMIND

Survival first. Profit second. Every decision auditable.